BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Subspaces

Interior, Exterior, Boundary and Subspaces

FEBRUARY 17, 2026

CONDUCTED BY

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

For updates subscribe on @ParthoSutraDhor

Interior, Exterior, Frontier and Boundary Points

Interior Point

Let A be a subset of a metric space (X, d). A point $a \in A$ is called an *interior point* of A if there exists $r > 0$ such that

$$a \in S_r(a) \subseteq A.$$

Interior of a Set

The set of all interior points of A is called the *interior* of A and is denoted by $\operatorname{int} A$, where

$$\operatorname{int} A = \{x \in A : S_r(x) \subseteq A \text{ for some } r > 0\}.$$

Clearly, $\operatorname{int} A$ is an open subset of A.

Exterior Point

A point $x \in X$ is called an *exterior point* of A if it is an interior point of the complement of A, that is, if there exists $r > 0$ such that

$$S_r(x) \subseteq A^c \quad \text{or equivalently} \quad S_r(x) \cap A = \emptyset.$$

Exterior of a Set

The set of all exterior points of A, denoted by $\operatorname{ext} A$, is called the *exterior* of A. By definition,

$$\operatorname{ext} A = \operatorname{int} A^c.$$

Hence $\operatorname{ext} A$ is open and is the largest open set contained in A^c. Moreover,

$$\operatorname{ext} A = (\overline{A})^c.$$

Boundary Point

A point $x \in X$ is called a *boundary point* of A if every open sphere centered at x intersects both A and its complement A^c, that is, for every $r > 0$,

$$S_r(x) \cap A \neq \emptyset \quad \text{and} \quad S_r(x) \cap A^c \neq \emptyset.$$

Boundary of a Set

The set of all boundary points of A is called the *boundary* of A and is denoted by ∂A, where

$$\partial A = \overline{A} \setminus \operatorname{int} A.$$

Equivalently,

$$\partial A = \overline{A} \cap \overline{A^c}.$$

Illustrations of Interior, Exterior, Boundary

Compute the interior, exterior, and boundary of the following subsets.

1. Let $X = \mathbb{R}$ with the usual metric and let $A = \mathbb{Q}$. Then

$$\operatorname{int} A = \emptyset, \qquad \operatorname{ext} A = \emptyset, \qquad \partial A = \mathbb{R}.$$

2. Let $X = \mathbb{R}$ with the usual metric and let $A = (1,2) \cup (3,4)$. Then

$$\operatorname{int} A = A,$$

$$\operatorname{ext} A = (-\infty, 1) \cup (2,3) \cup (4, \infty),$$

$$\partial A = \{1,2,3,4\}.$$

3. If (X,d) is a discrete metric space and $A \subseteq X$, then

$$\operatorname{int} A = A, \qquad \operatorname{ext} A = A^c, \qquad \partial A = \emptyset.$$

4. Let $X = \mathbb{N}$ with the usual metric and let $A = \{1,2,3\}$. Then

$$\operatorname{int} A = \emptyset, \qquad \operatorname{ext} A = \emptyset, \qquad \partial A = \mathbb{N}.$$

Properties of Interior

Let A and B be any two subsets of a metric space (X, d). Then:

1. $\operatorname{int} A$ is the largest open set contained in A, that is,

$$\operatorname{int} A = \bigcup\{G : G \text{ is open and } G \subseteq A\};$$

2. A is open if and only if $A = \operatorname{int} A$;
3. $A \subseteq B$ implies $\operatorname{int} A \subseteq \operatorname{int} B$;
4. $\operatorname{int}(A \cap B) = \operatorname{int} A \cap \operatorname{int} B$;
5. $\operatorname{int}(A \cup B) \supseteq \operatorname{int} A \cup \operatorname{int} B$.

Properties of Exterior of a Set

Let (X, d) be a metric space and let A, B be any two subsets of X. Then:

1. $\operatorname{ext} A$ is the largest open set contained in A^c;
2. A is closed if and only if $A^c = \operatorname{ext} A$;
3. $A \subseteq B$ implies $\operatorname{ext} B \subseteq \operatorname{ext} A$;
4. $\operatorname{ext}(A \cap B) \supseteq \operatorname{ext} A \cup \operatorname{ext} B$;
5. $\operatorname{ext}(A \cup B) = \operatorname{ext} A \cap \operatorname{ext} B$.

Properties of Boundary

Let (X,d) be a metric space, and let A,B be subsets of X. Then:

1. $\partial A = \overline{A} \cap \overline{A^c} = \overline{A} \setminus \operatorname{int} A$;
2. $\partial A = \emptyset$ if and only if A is both open and closed;
3. A is closed if and only if $\partial A \subseteq A$;
4. A is open if and only if $\partial A \subseteq A^c$;
5. $\partial(A \cap B) \subseteq \partial A \cup \partial B$. The equality holds if $\overline{A} \cap \overline{B} = \emptyset$;
6. $\partial(\operatorname{int} A) \subseteq \partial A$.

Subspaces

Subspace and Induced Metric

Let (X, d) be a metric space and let Y be a non-empty subset of X. The restriction of the metric d to $Y \times Y$, denoted by d_Y, is a metric on Y called the *induced metric*. The metric space (Y, d_Y) is called a *subspace* of (X, d).

Examples of Subspaces

Verify that:

1. the closed unit interval $[0, 1]$ and the set of all rational numbers $\mathbb{Q}$ are subspaces of $\mathbb{R}$ with the usual metric;
2. the unit circle, the closed unit disc, and the open unit disc are subspaces of the metric space $(\mathbb{C}, d)$ of complex numbers;
3. the real line $\mathbb{R}$ itself is a subspace of the complex plane $\mathbb{C}$.

Open Sphere in a Subspace

Let (X, d) be a metric space, let $Y \subseteq X$, and let $y \in Y$. The open sphere in the subspace (Y, d_Y) centered at y with radius $r > 0$ is denoted by $S_r^Y(y)$ and is defined by

$$S_r^Y(y) = \{x \in Y : d_Y(x, y) < r\}.$$

Relation Between Open Spheres

Show that for every $y \in Y$ and every $r > 0$,

$$S_r^Y(y) = S_r(y) \cap Y,$$

where $S_r(y)$ denotes the open sphere in (X, d).

Open Sets in Subspaces

Let (X, d) be a metric space and let $Y \subseteq X$. A subset A of Y is open in (Y, d_Y) if and only if there exists a set G open in (X, d) such that

$$A = G \cap Y.$$

Open Sets in a Subspace vs Ambient Space

Show that if $Y \subseteq X$ and a subset of Y is open in (X, d), then it is also open in the subspace (Y, d_Y). Show by examples that the converse need not be true.

Open in a Subspace may Not be open in Superspace

Let $X = \mathbb{R}$ with the usual metric and let

$$Y = [0, 1].$$

Then the open sphere in (Y, d_Y) centered at 0 with radius $\frac{1}{2}$ is

$$S^Y_{1/2}(0) = [0, \tfrac{1}{2}),$$

which is open in Y but not open in $\mathbb{R}$. Note that Y itself is not open in $\mathbb{R}$.

Example: Real Line vs Complex Plane

Show that an open interval of the real line is not an open subset of the complex plane $\mathbb{C}$ with the usual metric.

Thank You!

We'd love your questions and feedback.

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

@ParthoSutraDhor

(Lectures, walkthroughs, and course updates)

Scan for the channel

References

[1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.

[2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.